

SKIN DISEASE CLASSIFICATION USING TRANSFER LEARNING ON DERMOSCOPY IMAGES

Authors: SORIANO, ELENA F. | CABRERA, MARK R.

Year: 2023

Department of Computer Science

Degree Program: BSCS

Keywords: transfer learning, skin disease, dermoscopy, EfficientNet, medical imaging

ABSTRACT

This study applies EfficientNet-B3 transfer learning for 7-class skin disease classification on the ISIC 2019 dataset, achieving 89.1% accuracy and 0.91 AUC-ROC, matching dermatologist-level performance on common conditions.

INTRODUCTION

Skin cancer is the most common cancer globally, and early diagnosis significantly improves survival rates. Dermatologist shortages in rural Philippine provinces make early detection challenging. Computer-aided diagnosis using deep learning on dermoscopy images can extend diagnostic coverage. This study fine-tunes EfficientNet-B3 on the ISIC 2019 benchmark dataset for 7-class skin lesion classification.

METHODOLOGY

EfficientNet-B3 pretrained on ImageNet was fine-tuned on ISIC 2019 (25,331 dermoscopy images, 7 classes). Class imbalance addressed by weighted cross-entropy + oversampling of minority classes. Training: 30 epochs, cosine LR schedule (initial LR=1e-4), batch size 32, image size 300×300. Train/val/test: 70/15/15 stratified split. Augmentation: elastic distortion, cutout, color jitter. Evaluation: accuracy, per-class AUC, confusion matrix.

RESULTS

EfficientNet-B3 achieved 89.1% accuracy and mean AUC-ROC of 0.91 on the test set. Melanoma class AUC was 0.93. Dermatofibroma class showed lowest performance (AUC 0.85) due to visual similarity to nevi. Number of parameters: 12M. Inference time: 18ms per image on NVIDIA RTX 3060.

DISCUSSION

EfficientNet-B3 delivers clinically relevant performance for automated skin lesion classification. The 0.93 melanoma AUC is comparable to published dermatologist performance (0.91-0.94). The model's confidence calibration is slightly overconfident on minority classes. Future work should validate on a locally-collected Filipino skin lesion dataset to address skin tone distribution differences.